

Instructions:

Max. Points: 100

- 1- This is hand written assignment. You have to submit the hard copy on 31st October 2025 in your class as well scan copy on Google classroom.
- 2- Just write the question number instead of writing the whole question.
- 3- You can only use A4 size paper for solving the assignment.
- 4- You have to write your student ID and Section on the top of each page.

1. Let R be the following relation defined on the set {a, b, c, d}:

$$R = \{(a, a), (a, c), (a, d), (b, a), (b, b), (b, c), (b, d), (c, b), (c, c), (d, b), (d, d)\}.$$

Determine whether R is:

- a) Reflexive b) Symmetric c) Antisymmetric d) Transitive e) Irreflexive f) Asymmetric

Solution:

- (a) R is reflexive because R contains (a, a), (b, b), (c, c), and (d, d).
 (b) R is not symmetric because R contains (a, c) but not (c, a) ∈ R.
 (c) R is not antisymmetric because both (b, c) ∈ R and (c, b) ∈ R, but b ≠ c.
 (d) R is not Transitive because both (a, c) ∈ R and (c, b) ∈ R, but not (a, b) ∈ R.
 (e) R is not irreflexive because R contains (a, a), (b, b), (c, c), and (d, d).
 (f) R is not Asymmetric because R is not Antisymmetric.

2. List the ordered pairs in the relation R from A = {0, 1, 2, 3, 4} to B = {0, 1, 2, 3}, where (a, b) ∈ R if and only if
 a) a = b. b) a + b = 4. c) a > b. d) a | b. e) gcd(a, b) = 1. f) lcm(a, b) = 2.

Solution:

- a) { (0,0), (1, 1), (2, 2), (3, 3) }
 b) { (1, 3), (2, 2), (3, 1), (4, 0) }
 c) { (1, 0), (2, 0), (3, 0), (4, 0), (2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3) }
 d) { (1, 0), (2, 0), (3, 0), (4, 0), (1, 1), (1,2), (2,2), (1,3), (3,3) }
 e) { (1,0), (0,1), (1,1), (1,2), (1,3), (2,1), (3,1), (4,1), (2,3), (3,2), (4,3) }
 f) { (1,2), (2,1), (2,2) }

3. List all the ordered pairs in the relation R = {(a, b) | a divides b} on the set {1, 2, 3, 4, 5, 6}.

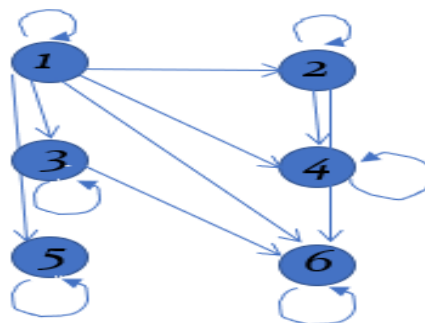
Display this relation as Directed Graph (digraph), as well in matrix form.

Determine whether the relation R is Equivalence or Partial Order Relation.

Solution:

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$



The relation is not an equivalence relation since it does not hold symmetric property.

The relation is partial order relation since it holds Reflexive, Antisymmetric and Transitive property.

4. For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$

b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

c) $\{(2, 4), (4, 2)\}$

d) $\{(1, 2), (2, 3), (3, 4)\}$

e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$

f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$

Solution:

(a) R is not reflexive: It doesn't contain $(1,1)$ and $(4,4)$.

(b) R is not symmetric because R contains $(2, 4)$ but not $(4, 2) \in R$.

(c) R is not antisymmetric: we have $(2,3)$ and $(3,2)$ but $2 \neq 3$.

(d) R is Transitive because for any numbers a, b, and c, if $(a, b), (b, c) \in R$ then $(a, c) \in R$.

b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

Solution:

(a) R is reflexive: It contains $(1,1), (2,2), (3,3)$ and $(4,4)$.

(b) R is symmetric because (a,b) and $(b,a) \in R$.

(c) R is not antisymmetric: we have $(1,2)$ and $(2,1)$ but $1 \neq 2$.

(d) R is Transitive because for any numbers a, b, and c, if $(a, b), (b, c) \in R$ then $(a, c) \in R$.

c) $\{(2, 4), (4, 2)\}$

Solution:

(a) R is not reflexive: It doesn't contain $(1,1), (2,2), (3,3)$ and $(4,4)$.

(b) R is symmetric because R contains $(2, 4)$ and $(4, 2) \in R$.

(c) R is not antisymmetric: we have $(2,4)$ and $(4,2)$ but $2 \neq 4$.

(d) R is not Transitive because $(2,4), (4, 2) \in R$ but not $(2,2) \in R$.

d) $\{(1, 2), (2, 3), (3, 4)\}$

Solution:

(a) R is not reflexive: It doesn't contain $(1,1), (2,2), (3,3)$ and $(4,4)$.

(b) R is not symmetric because $(1,2) \in R$ but not $(2,1) \in R$.

(c) R is antisymmetric: we have (a,b) but not $(b,a) \in R$.

(d) R is not Transitive because $(1,2), (2, 3) \in R$ but not $(1,3) \in R$.

e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$

Solution:

(a) R is reflexive: It contains $(1,1), (2,2), (3,3)$ and $(4,4)$.

(b) R is symmetric because R contains (a,b) and $(b,a) \in R$.

(c) R is antisymmetric: we have (a,b) and $(b,a) \in R$ then $a = b$.

(d) R is Transitive because for any numbers a, b, and c, if $(a, b), (b, c) \in R$ then $(a, c) \in R$.

f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

Solution:

(a) R is not reflexive: It doesn't contain $(1,1), (2,2), (3,3)$ and $(4,4)$.

(b) R is not symmetric because $(1,4) \in R$ but not $(4,1) \in R$.

(c) R is not antisymmetric: we have $(1,3)$ and $(3,1) \in R$ but $1 \neq 3$.

(d) R is not Transitive because we have $(1,3)$ and $(3,1) \in R$ but not $(1,1) \in R$.

5. Determine whether the relation R on the set of all people is reflexive, symmetric, antisymmetric, Asymmetric, irreflexive and/or transitive, where $(a, b) \in R$ if and only if:

a) a is taller than b.

Solution: The relation R is **not reflexive**, because a person cannot be taller than himself/herself.

The relation R is **not symmetric**, because if person A is taller than person B, then person B is NOT taller than person A.

The relation R is **antisymmetric**, because $(a, b) \in R$ and $(b, a) \in R$ cannot occur at the same time (as one person is always taller than the other, but not the other way around).

The relation R is **transitive**, because if person A is taller than person B and if person B is taller than person C, then person A needs to be taller than person C as well.

b) a and b
born on
same day.

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the

Solution:

The relation R is **reflexive**, because a person is born on the same day as himself/herself.

The relation R is **symmetric**, because if person A and person B are born on the same day, then person B is also born on the same day as person A.

The relation R is **not antisymmetric**, because if person A and person B are born on the same day and if person B and person A are born on the same day, then these two people are not necessarily the same person.

The relation R is **transitive**, because if person A and person B are born on the same day and if person B and person C are born on the same day, then person A and person C are also born on the same day.

c) a has the same first name as b.

Solution:

The relation R is **reflexive**, because a person has the same first name as himself/herself.

The relation R is **symmetric**, because if person A has the same first name as person B, then person B also has the same first name as person A.

The relation R is **not antisymmetric**, because if person A has the same first name as person B and if person B also has the same first name as person A, then these two people are not necessarily the same person (as there are different people with the same first name).

The relation R is **transitive**, because if person A has the same first name as person B and if person B also has the same first name as person C, then person A also has the same first name as person C.

d) a and b have a common grandparent.

Solution:

The relation R is **reflexive**, because a person has the same grandparents as himself/herself.

The relation R is **symmetric**, because if person A and person B have a common grandparent, then person B and person A also have a common grandparent.

The relation R is **not antisymmetric**, because if person A and person B have a common grandparent and if person B and person A have a common grandparent, then these two people are not necessarily the same person (as there are different people with the same grandparents).

The relation R is **not transitive**, because if person A and person B have a common grandparent and if person B and person C have a common grandparent, then person A and person C do not necessarily have a common grandparent (for example, the common grandparent of A and B can be from person B's father's side of the family, while the common grandparent of B and C can be from person B's mother's side of the family).

6. Give an example of a relation on a set of Whole Numbers that is:

a) both symmetric and antisymmetric.

Solution: $\{(1,1), (2,2), (3,3), (4,4)\}$

b) neither symmetric nor antisymmetric.

Solution: $\{(1,2), (2,1), (3,4)\}$

c) both symmetric and transitive

Solution: $\{(1,2), (2,1), (1,1)\}$

d) neither Symmetric nor transitive

Solution: $\{(1,2), (2,1), (3,4)\}$

7. Consider these relations on the set of real numbers: $A = \{1,2,3\}$

$R1 = \{(a, b) \in R \mid a > b\}$, the "greater than" relation,

$R2 = \{(a, b) \in R \mid a \geq b\}$, the "greater than or equal to" relation,

$R3 = \{(a, b) \in R \mid a < b\}$, the "less than" relation,

$R4 = \{(a, b) \in R \mid a \leq b\}$, the "less than or equal to" relation,

$R5 = \{(a, b) \in R \mid a = b\}$, the "equal to" relation,

$R6 = \{(a, b) \in R \mid a \neq b\}$, the "unequal to" relation.

Find:

a) $R2 \cup R4$.

b) $R3 \cup R6$.

c) $R3 \cap R6$.

d) $R4 \cap R6$.

e) $R3 - R6$.

f) $R6 - R3$.

g) $R2 \oplus R6$.

h) $R3 \oplus R5$.

i) $R2 \circ R1$.

j) $R6 \circ R6$.

Solution:

$R1 = \{(2,1), (3,1), (3,2)\}$

$R2 = \{(1,1), (2,2), (3,3), (2,1), (3,1), (3,2)\}$

$R3 = \{(1,2), (1,3), (2,3)\}$

$R4 = \{(1,1), (2,2), (3,3), (1,2), (1,3), (2,3)\}$

$R5 = \{(1,1), (2,2), (3,3)\}$

$R6 = \{(1,2), (1,3), (2,1), (2,3), (3,1), (3,2)\}$

a) $R2 \cup R4 = \{(1,1), (2,2), (3,3), (2,1), (3,1), (3,2), (1,2), (1,3), (2,3)\}$

b) $R3 \cup R6 = \{(1,2), (1,3), (2,1), (2,3), (3,1), (3,2)\}$

c) $R3 \cap R6 = \{(1,2), (1,3), (2,3)\}$

d) $R4 \cap R6 = \{(1,2), (1,3), (2,3)\}$

e) $R3 - R6 = \{\}$ OR Φ

f) $R6 - R3 = \{(2,1), (3,1), (3,2)\}$

g) $R2 \oplus R6 = \{(1,1), (2,2), (3,3), (1,2), (1,3), (2,3)\}$

h) $R3 \oplus R5 = \{(1,1), (2,2), (3,3), (1,2), (1,3), (2,3)\}$

i) $R2 \circ R1 = \{(2,1), (3,1), (3,2)\}$

j) $R6 \circ R6 = \{(1,1), (2,2), (3,3), (2,1), (3,1), (3,2), (1,2), (1,3), (2,3)\}$

8. a) Represent each of these relations on $\{1, 2, 3\}$ with a matrix (with the elements of this set listed in increasing order).

i) $\{(1, 1), (1, 2), (1, 3)\}$

ii) $\{(1, 2), (2, 1), (2, 2), (3, 3)\}$

iii) $\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

iv) $\{(1, 3), (3, 1)\}$

i) $\{(1, 1), (1, 2), (1, 3)\}$

Solution:
$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

ii) $\{(1, 2), (2, 1), (2, 2), (3, 3)\}$

Solution:
$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

iii) $\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

Solution:
$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

iv) $\{(1, 3), (3, 1)\}$

Solution: $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

b) List the ordered pairs in the relations on $\{1, 2, 3\}$ corresponding to these matrices (where rows and columns correspond to the integers listed in increasing order).

(i) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ Solution: $R = \{ (1,1), (1,3), (2,2), (3,1), (3,3) \}$

(ii) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ Solution: $R = \{ (1,2), (2,2), (3,2) \}$

(iii) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ Solution: $R = \{ (1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3) \}$

9. a) Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x . Is R an equivalence relation?

Solution:

Show that all of the properties of an equivalence relation hold.

- Reflexivity: Because $l(a) = l(a)$, it follows that aRa for all strings a .
- Symmetry: Suppose that aRb . Since $l(a) = l(b)$, $l(b) = l(a)$ also holds and bRa .
- Transitivity: Suppose that aRb and bRc . Since $l(a) = l(b)$, and $l(b) = l(c)$, $l(a) = l(c)$ also holds and aRc .

b) Let m be an integer with $m > 1$. Show that the relation $R = \{ (a,b) \mid a \equiv b \pmod{m} \}$ is an equivalence relation on the set of integers.

Solution:

Recall that $a \equiv b \pmod{m}$ if and only if m divides $a - b$.

- Reflexivity: $a \equiv a \pmod{m}$ since $a - a = 0$ is divisible by m since $0 = 0 \cdot m$.
- Symmetry: Suppose that $a \equiv b \pmod{m}$. Then $a - b$ is divisible by m , and so $a - b = km$, where k is an integer. It follows that $b - a = (-k)m$, so $b \equiv a \pmod{m}$.
- Transitivity: Suppose that $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. Then m divides both $a - b$ and $b - c$. Hence, there are integers k and l with $a - b = km$ and $b - c = lm$. We obtain by adding the equations: $a - c = (a - b) + (b - c) = km + lm = (k + l)m$. Therefore, $a \equiv c \pmod{m}$.

10. a) Find the first five terms of the sequence for each of the following general terms where $n > 0$.

(i) $2^n - 1$

Solution:

1, 3, 7, 15, 31 are the first five terms of the given sequence.

(ii) $10 - \frac{3}{2}n$

Solution:

$\frac{17}{2}, 7, \frac{11}{2}, 4, \frac{5}{2}$ are the first five terms.

(iii) $\frac{(-1)^n}{n^2}$

Solution:

$-1, \frac{1}{4}, -\frac{1}{9}, \frac{1}{16}, -\frac{1}{25}$ are the first five terms.

(iv) $\frac{3n+4}{2n-1}$

Solution: $7, \frac{10}{3}, \frac{13}{5}, \frac{16}{7}, \frac{19}{9}$ are the first five terms.

b) Identify the following Sequence as Arithmetic or Geometric Sequence then find the indicated term.

(i) -15, -22, -29, -36,; 11th term.

Solution:

Here common difference (d) = -7

$$T_n = a + (n - 1)d; T_{11} = -15 + (11 - 1)(-7) = -85$$

(ii) a - 42b, a - 39b, a - 36b, a - 33b, ..., 15th term.

Solution:

Here common difference (d) = 3b

$$T_n = a + (n - 1)d; \quad T_{15} = a - 42b + (15 - 1)(3b) = a$$

(iii) 4, 3, $\frac{9}{4}$, ...; 17th term

Solution:

Here common ratio (r) = $\frac{3}{4}$

$$T_n = ar^{n-1}; \quad T_{17} = 4\left(\frac{3}{4}\right)^{17-1} = \frac{3^{16}}{4^{15}}$$

(iv) 32, 16, 8, ..., 9th term

Solution:

Here common ratio (r) = $\frac{1}{2}$

$$T_n = ar^{n-1}; \quad T_{17} = 32\left(\frac{1}{2}\right)^{9-1} = \frac{1}{8}$$

11. a) Find the G.P in which:

(i) $T_3 = 10$ and $T_5 = 2 \frac{1}{2}$

Solution:

Since $T_n = ar^{n-1}$

$$T_3 = ar^2 = 10 \text{ ----(i)} \quad T_5 = ar^4 = \frac{5}{2} \text{ ----(ii)}$$

Now, dividing (ii) by dividing (i), we get $r = \pm \frac{1}{2}$ and putting it in (i) we get $a = 40$.

Now the required G.P is 40, 20, 10, 5, $\frac{5}{2}$, ... OR 40, -20, 10, -5, $\frac{5}{2}$,

(ii) $T_5 = 8$ and $T_8 = -\frac{64}{27}$

Solution:

Since $T_n = ar^{n-1}$

$$T_5 = ar^4 = 8 \text{ ---- (i)} \quad T_8 = ar^7 = -\frac{64}{27} \text{ ----(ii)}$$

Now, dividing (ii) by dividing (i) we get $r = -\frac{2}{3}$ and putting it in (i) we get $a = \frac{81}{2}$.

Now the required G.P is $\frac{81}{2}$, -27, 18, -12, 8, ...

(b) Find the A.P in which:

(i) $T_4 = 7$ and $T_{16} = 31$

Solution:

Since $T_n = a + (n-1)d$;

$$T_4 = a + 3d = 7 \text{(i)} \quad T_{16} = a + 15d = 31 \text{(ii)}$$

Now subtracting (ii) from (i), we get $d = 2$ and putting it in (i) we get $a = 1$.

Now the required A.P is 1, 3, 5, 7, 9, 11, ...

(ii) $T_5 = 86$ and $T_{10} = 146$

Solution:

Since $T_n = a + (n-1)d$;

$$T_5 = a + 4d = 86 \text{(i)} \quad T_{10} = a + 9d = 146 \text{(ii)}$$

Now subtracting (ii) from (i), we get $d = 12$ and putting it in (i) we get $a = 38$.

Now the required A.P is 38, 50, 62, 74, 86, ...

12. a) How many numbers are there between 256 and 789 that are divisible by 7? Also find their sum.

Solution:

First, we find the A.P with the common difference (d) = 7

259, 266, 273, 280, 784

$$\text{Since } T_n = a + (n-1)d; \quad 784 = 259 + (n-1)(7); \quad n = 76.$$

Now for Sum; $S_n = \frac{n}{2} [2a + (n-1)d]$;
 $S_{76} = \frac{76}{2} [2(259) + (76-1)(7)] = 39,634$.

- b) Find the sum to n terms of an A.P whose first term is $\frac{1}{n}$ and the last term is $\frac{n^2-n+1}{n}$.

Solution:

Since, $S_n = \frac{n}{2} [2a + (n-1)d]$ ----- (i)

1st we have to find "d"

Now, $T_n = a + (n-1)d$

$$\frac{n^2 - n + 1}{n} = \frac{1}{n} + (n-1)d$$

Finally, $d = 1$. Hence putting it in we get,

$$S_n = \frac{n^2 - n + 1}{2}.$$

13. a) Use summation notation to express the sum of the first 100 terms of the sequence $\{a_j\}$, where $a_j = \frac{1}{j}$ for $j = 1, 2, 3, \dots$

Solution:

The lower limit for the index of summation is 1, and the upper limit is 100. We write this sum as $\sum_{j=1}^{100} \frac{1}{j}$.

- b) What is the value of:

(i) $\sum_{k=4}^8 (-1)^k$.

Solution:

$$= (-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8$$

$$= 1 + (-1) + 1 + (-1) + 1 = 1.$$

(ii) $\sum_{j=1}^5 (j)^2$.

Solution:

$$= 1^2 + 2^2 + 3^2 + 4^2 + 5^2$$

$$= 1 + 4 + 9 + 16 + 25 = 55.$$

14. a) A batsman hits the ball in the air after reaching the maximum height, ball dropped 40 feet during the 1st Second, 33 feet during the 2nd Second, and 26 feet during the 3rd Second, and so on. How many feet the ball dropped in 5th Second. Also find the total distance covered by the ball after 5th second.

Solution:

Here $a = 40$, $d = T_2 - T_1 = 33 - 40 = -7$, $n = 5$, $T_5 = ?$

$$T_n = a + (n-1)d; \quad T_5 = 40 + (5-1)(-7) = 12.$$

The ball will fall 12 feet in 5th Second.

$$\text{Now, } S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_5 = \frac{5}{2} [2(40) + (5-1)(-7)]$$

$$S_5 = 130. \text{ Hence ball covered 130 feet.}$$

- b) If the population of a town increases geometrically at the rate of 7% annually and the present population is 70000. What will be the population after 6 years from now?

Solution:

Present population = $a = 70000$; Number of years = $n = 6$;

Number of terms = $n = 7$

Rate of increase of population = $r = 1 + 7\% = 1.07$

Population after 6 years = $T_7 = ?$

$$T_n = ar^{n-1} = 70000(1.07)^{7-1} = 105051 \text{ approx.}$$

The population after 7 years will be 105051.

15. a) Find the first six terms of the sequence defined by each of these recurrence relations and initial conditions.

i) $a_n = -2a_{n-1}$, $a_0 = -1$

Solution:

$$\begin{aligned}
a_0 &= -1 \\
a_1 &= -2a_0 = -2(-1) = 2 \\
a_2 &= -2a_1 = -2(2) = -4 \\
a_3 &= -2a_2 = -2(-4) = 8 \\
a_4 &= -2a_3 = -2(8) = -16 \\
a_5 &= -2a_4 = -2(-16) = 32
\end{aligned}$$

ii) $a_n = a_{n-1} - a_{n-2}$, $a_0 = 2$, $a_1 = -1$

Solution:

$$\begin{aligned}
a_0 &= 2 \\
a_1 &= -1 \\
a_2 &= a_1 - a_0 = -1 - 2 = -3 \\
a_3 &= a_2 - a_1 = -3 - (-1) = -2 \\
a_4 &= a_3 - a_2 = -2 - (-3) = 1 \\
a_5 &= a_4 - a_3 = 1 - (-2) = 3
\end{aligned}$$

iii) $a_n = 3a_{n-1}^2$, $a_0 = 1$

Solution:

$$\begin{aligned}
a_0 &= 1 \\
a_1 &= 3a_0^2 = 3(1^2) = 3 \\
a_2 &= 3a_1^2 = 3(3^2) = 3(9) = 27 \\
a_3 &= 3a_2^2 = 3(27^2) = 3(729) = 2187 \\
a_4 &= 3a_3^2 = 3(2187^2) = 3(4782969) = 14348907 \\
a_5 &= 3a_4^2 = 3(14348907^2) = 3(205891132094649) = 617673396283947
\end{aligned}$$

iv) $a_n = na_{n-1} + a_{n-2}^2$, $a_0 = -1$, $a_1 = 0$

Solution:

$$\begin{aligned}
a_0 &= -1 \\
a_1 &= 0 \\
a_2 &= 2a_1 + a_0^2 = 2(0) + (-1)^2 = 0 + 1 = 1 \\
a_3 &= 3a_2 + a_1^2 = 3(1) + 0^2 = 3 + 0 = 3 \\
a_4 &= 4a_3 + a_2^2 = 4(3) + 1^2 = 12 + 1 = 13 \\
a_5 &= 5a_4 + a_3^2 = 5(13) + 3^2 = 65 + 9 = 74
\end{aligned}$$

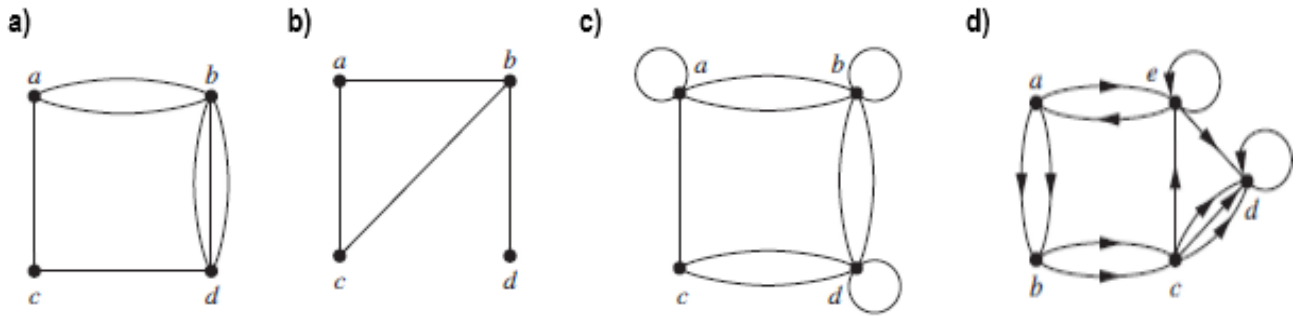
b) What are the terms a_0 , a_1 , a_2 and a_3 of the sequence $\{a_n\}$, where a_n equals?

- i) $(-2)^n$ ii) 3 iii) $7 + 4^n$ iv) $2^n + (-2)^n$

Solution:

- i) $a_0 = (-2)^0 = 1$, $a_1 = (-2)^1 = -2$, $a_2 = (-2)^2 = 4$, $a_3 = (-2)^3 = -8$
ii) $a_0 = a_1 = a_2 = a_3 = 3$
iii) $a_0 = 7 + 4^0 = 8$, $a_1 = 7 + 4^1 = 11$, $a_2 = 7 + 4^2 = 23$, $a_3 = 7 + 4^3 = 71$
iv) $a_0 = 2^0 + (-2)^0 = 2$, $a_1 = 2^1 + (-2)^1 = 0$, $a_2 = 2^2 + (-2)^2 = 8$, $a_3 = 2^3 + (-2)^3 = 0$

16. Determine whether the graph shown in figure i to iv has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops. Use your answers to determine the type of graph.



Solution:

a) It has undirected edges.

It has multiple edges.

It has no loops.

It is undirected Multigraph.

b) It has undirected edges.

It has no multiple edges.

It has no loops.

It is undirected simple graph.

c) It has undirected edges.

It has multiple edges.

It has three loops.

It is undirected Pseudo graph.

d) It has directed edges.

It has multiple edges.

It has two loops.

It is directed Multi graph.

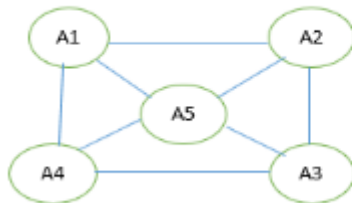
17. The intersection graph of a collection of sets $A_1, A_2, \dots, A_n$ is the graph that has a vertex for each of these sets and has an edge connecting the vertices representing two sets if these sets have a nonempty intersection. Construct the intersection graph of these collections of sets.

a) $A_1 = \{0, 2, 4, 6, 8\}$, $A_2 = \{0, 1, 2, 3, 4\}$, $A_3 = \{1, 3, 5, 7, 9\}$, $A_4 = \{5, 6, 7, 8, 9\}$, $A_5 = \{0, 1, 8, 9\}$

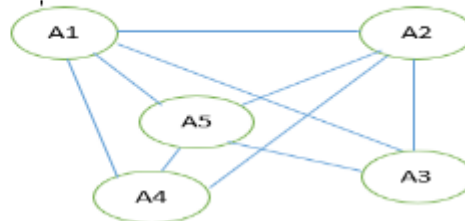
b) $A_1 = \{\dots, -4, -3, -2, -1, 0\}$, $A_2 = \{\dots, -2, -1, 0, 1, 2, \dots\}$, $A_3 = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}$, $A_4 = \{\dots, -5, -3, -1, 1, 3, 5, \dots\}$, $A_5 = \{\dots, -6, -3, 0, 3, 6, \dots\}$

Solution:

a)



b)



18. a) Find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Also find the neighborhood vertices of each vertex in given graphs.

Solution:

i) Number of Vertices: 5 Number of edges: 13

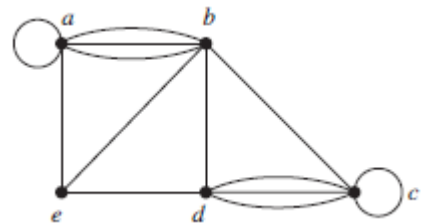
Degree of vertices:

$\deg(a) = \deg(b) = \deg(c) = 6$, $\deg(d) = 5$, $\deg(e) = 3$.

Neighborhood Vertices:

$N(a) = \{a, b, e\}$, $N(b) = \{a, c, d, e\}$, $N(c) = \{b, c, d\}$, $N(d) = \{b, c, e\}$,

$N(e) = \{a, b, d\}$



ii) Number of Vertices: 9 Number of edges: 12

Degree of vertices:

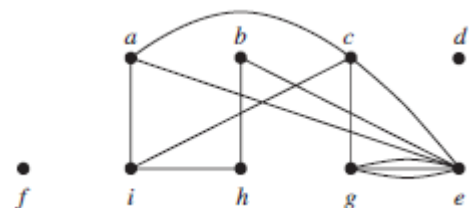
$\deg(a) = 3$, $\deg(b) = 2$, $\deg(c) = 4$, $\deg(d) = 0$, $\deg(e) = 6$.

$\deg(f) = 0$, $\deg(g) = 4$, $\deg(h) = 2$, $\deg(i) = 3$.

Neighborhood Vertices:

$N(a) = \{c, e, i\}$, $N(b) = \{e, h\}$, $N(c) = \{a, e, g, i\}$, $N(d) = \emptyset$,

$N(e) = \{a, b, c, g\}$, $N(f) = \emptyset$, $N(g) = \{c, e\}$, $N(h) = \{b, i\}$, $N(i) = \{a, c, h\}$.



b) Determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the given directed multigraph.

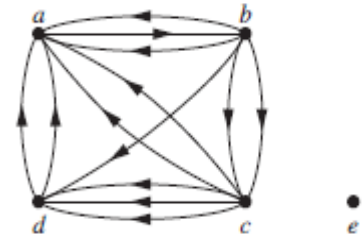
Solution:

i) In-degree of a vertices

$$\deg^-(a) = 6, \deg^-(b) = 1, \deg^-(c) = 2, \deg^-(d) = 4, \deg^-(e) = 0.$$

Out-degree of a vertices

$$\deg^+(a) = 1, \deg^+(b) = 5, \deg^+(c) = 5, \deg^+(d) = 2, \deg^+(e) = 0.$$



ii) In-degree of a vertices

$$\deg^-(a) = 2, \deg^-(b) = 3, \deg^-(c) = 2, \deg^-(d) = 1.$$

Out-degree of a vertices

$$\deg^+(a) = 2, \deg^+(b) = 4, \deg^+(c) = 1, \deg^+(d) = 1.$$

19. a) Suppose that a new company has five employees: Zamora, Agraharam, Smith, Chou, and Macintyre. Each employee will assume one of six responsibilities: planning, publicity, sales, marketing, development, and industry relations. Each employee is capable of doing one or more of these jobs: Zamora could do planning, sales, marketing, or industry relations; Agraharam could do planning or development; Smith could do publicity, sales, or industry relations; Chou could do planning, sales, or industry relations; and Macintyre could do planning, publicity, sales, or industry relations. Model the capabilities of these employees using appropriate graph.

Solution:

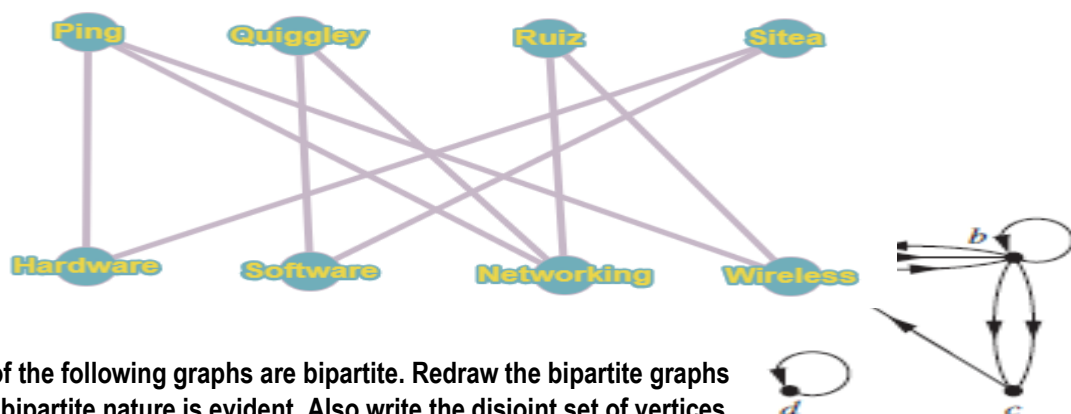
Bipartite Graph



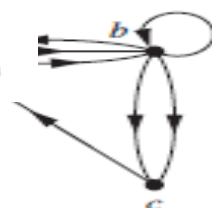
- b) Suppose that there are four employees in the computer support group of the School of Engineering of a large university. Each employee will be assigned to support one of four different areas: hardware, software, networking, and wireless. Suppose that Ping is qualified to support hardware, networking, and wireless; Quiggley is qualified to support software and networking; Ruiz is qualified to support networking and wireless, and Sitea is qualified to support hardware and software. Use appropriate graph to model the four employees and their qualifications.

Solution:

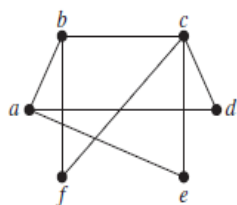
Bipartite Graph



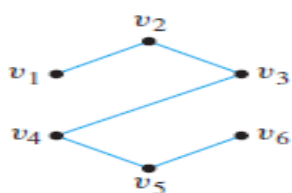
20. Find which of the following graphs are bipartite. Redraw the bipartite graphs so that their bipartite nature is evident. Also write the disjoint set of vertices.



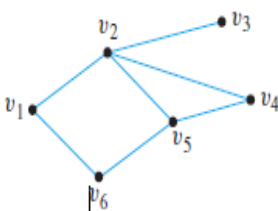
a)



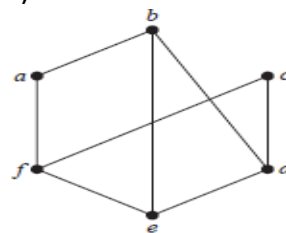
b)



c)



d)



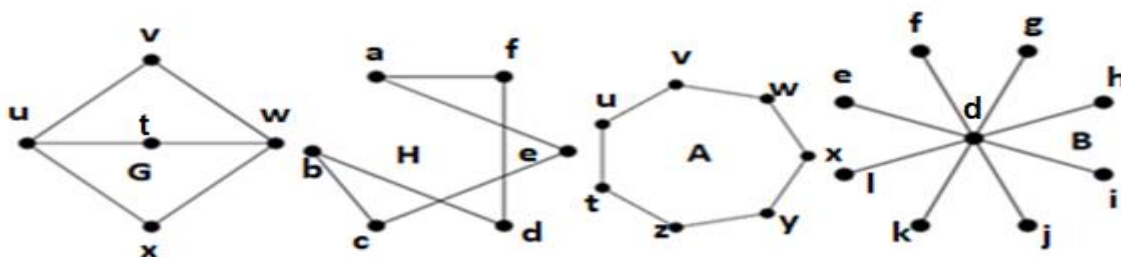
Solution:

a) Not bipartite (since a is adjacent to b & f vertices)

b) Bipartite (A {V₁, V₃, V₅} & B {V₂, V₄, V₆})c) Not bipartite (since V₄ & V₅ are adjacent vertices)

d) Not Bipartite (since b is adjacent to d & e vertices)

21. a) Which of the graphs below are bipartite? Justify your answers.



Solution:

Graph G is Bipartite since two set of disjoint vertices exists. A{v, t, x} and B{u, w}

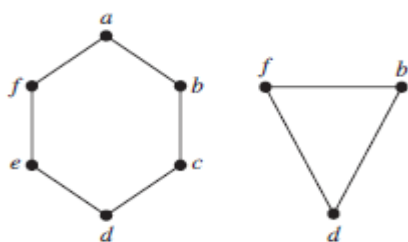
Graph H is Bipartite since two set of disjoint vertices exists. A{b, e, f} and B{a, c, d}

Graph A is not Bipartite since two set of disjoint vertices does not exists. Vertex y and z are adjacent.

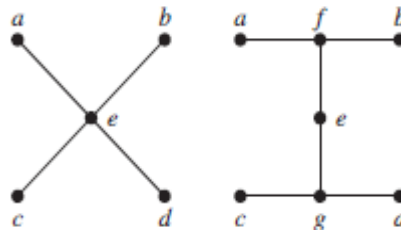
Graph B is Bipartite since two set of disjoint vertices exists. A{d} and B{e, f, g, h, i, j, k, l}

b) Find the union of the given pair of simple graphs. (Assume edges with the same endpoints are the same.)

i)

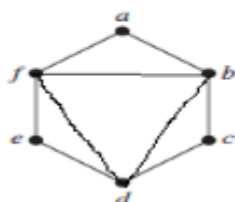


ii)

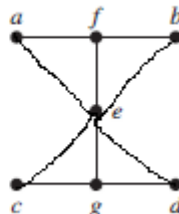


Solution:

i)



ii)



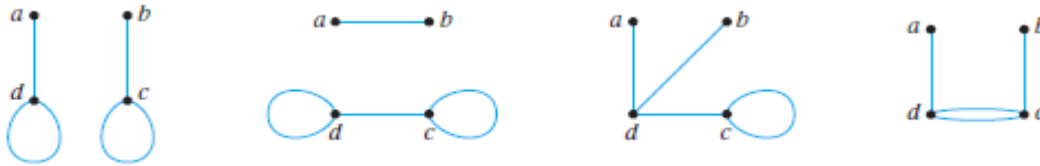
22. Draw a graph with the specified properties or show that no such graph exists.

Solution:

a) No such graph is possible. By Handshaking theorem, the total degree of a graph is even.

But a graph with four vertices of degrees 1, 1, 2, and 3 would have a total degree of $1 + 1 + 2 + 3 = 7$, which is odd.

b) Let G be any of the graphs shown below.



In each case, no matter how the edges are labeled, $\deg(a) = 1$, $\deg(b) = 1$, $\deg(c) = 3$, and $\deg(d) = 3$.

c) There is no simple graph with four vertices of degrees 1, 1, 3, and 3.

23. a) In a group of 15 people, is it possible for each person to have exactly 3 friends? Explain. (Assume that friendship is a symmetric relationship: If x is a friend of y , then y is a friend of x .)

Solution:

By using Handshaking theorem.

No! there is no graph possible, such that 15 vertices have degree 3. Since $(15 * 3) \neq 2e$.

b) In a group of 4 people, is it possible for each person to have exactly 3 friends? Why?

Solution:

By using Handshaking theorem.

Yes! there is graph possible, such that 4 vertices have degree 3. Since $(4 * 3) = 2e$.

c) If 10 people each shake hands with each other, how many handshakes took place? What does this question have to do with graph theory?

Solution:

This is asking for the number of edges in K_{10} . Each vertex (person) has degree (shook hands with) 9 (people). So the sum of the degrees is 90. However, the degrees count each edge (handshake) twice, so there are 45 edges in the graph. That is how many handshakes took place.

d) Among a group of 5 people, is it possible for everyone to be friends with exactly 2 of the people in the group? What about 3 of the people in the group?

Solution:

It is possible for everyone to be friends with exactly 2 people. You could arrange the 5 people in a circle and say that everyone is friends with the two people on either side of them (so you get the graph C_5). However, it is not possible for everyone to be friends with 3 people. That would lead to a graph with an odd number of odd degree vertices which is impossible since the sum of the degrees must be even.

e) How many vertices does a regular graph of degree four with 10 edges have?

Solution:

We want to determine a regular graph of degree four with $m = 10$ edges.

Let the graph contain n vertices $v_1, v_2, \dots, v_n$, then each of these n vertices have degree 4.

$$\deg(v_i) = 4$$

$$i = 1, 2, \dots, n$$

By the Handshaking theorem, the sum of degrees of all vertices is equal to twice the number of edges:

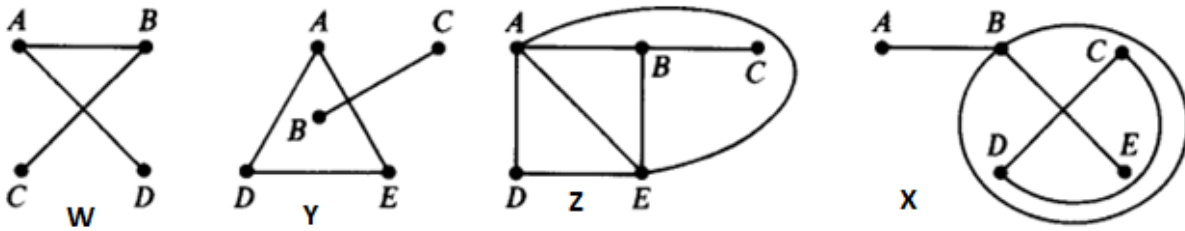
$$20 = 2(10) = 2m = \sum_{v=1}^n \deg(v_i) = \sum_{v=1}^n 4 = 4n$$

We then obtained the equation $20 = 4n$. Divide each side of the equation by 4:

$$n = \frac{20}{4} = 5$$

24. Consider the multigraphs and find the following.

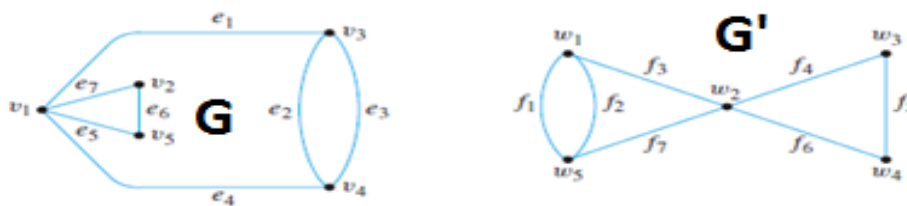
- Which of them are connected? If a graph is not connected, find its connected components.
- Which is/are cycle-free (without cycles)?
- Which is/are loop-free (without loops)?
- Which is/are (simple) graphs?



- Only (W) and (Z) are connected, (Y) is disconnected; its connected components are $\{A, D, E\}$ and $\{B, C\}$. (X) is disconnected; its connected components are $\{A, B, E\}$ and $\{C, D\}$.
- Only (W) and (X) are cycle-free. (Y) has the cycle (A, D, E, A) , and (Z) has the cycle (A, B, E, A) .
- Only (X) has a loop which is $\{B, B\}$.
- Only (W) and (Y) are graphs. Multigraph (Z) has multiple edges $\{A, E\}$ and $\{A, E\}$; and (X) has both multiple edges $\{C, D\}$ and $\{C, D\}$ and a loop $\{B, B\}$.

25. For given pair (G, G') of graphs. Determine whether they are isomorphic. If they are, give function $g: V(G) \rightarrow V(G')$ that define the isomorphism. If they are not, give an invariant for graph isomorphism that they do not share.

a)

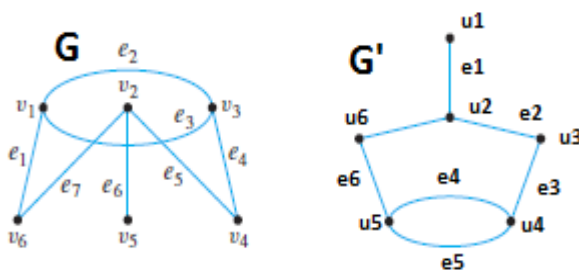


Solution:

Both graph G and G' are satisfying all the invariant. Hence, they are isomorphic.

Function: $g(v_1) = w_2$, $g(v_2) = w_3$, $g(v_3) = w_1$, $g(v_4) = w_5$, $g(v_5) = w_4$

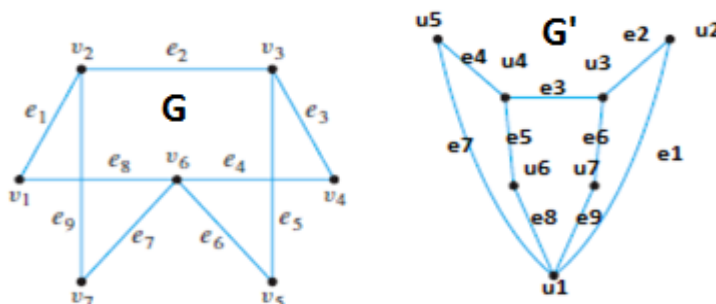
b)



Solution: Both graph G and G' are satisfying all the invariant. Hence, they are isomorphic.

Function: $g(v_1) = u_5$, $g(v_2) = u_2$, $g(v_3) = u_4$, $g(v_4) = u_3$, $g(v_5) = u_1$, $g(v_6) = u_6$

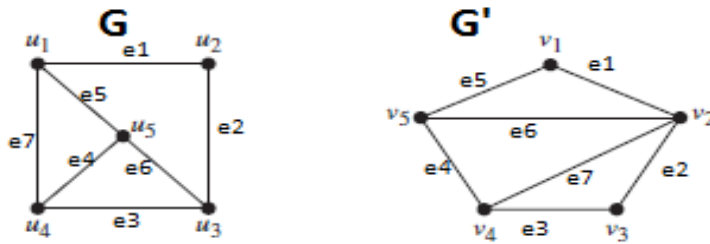
c)



Solution: Both graph G and G' are satisfying all the invariant. Hence, they are isomorphic.

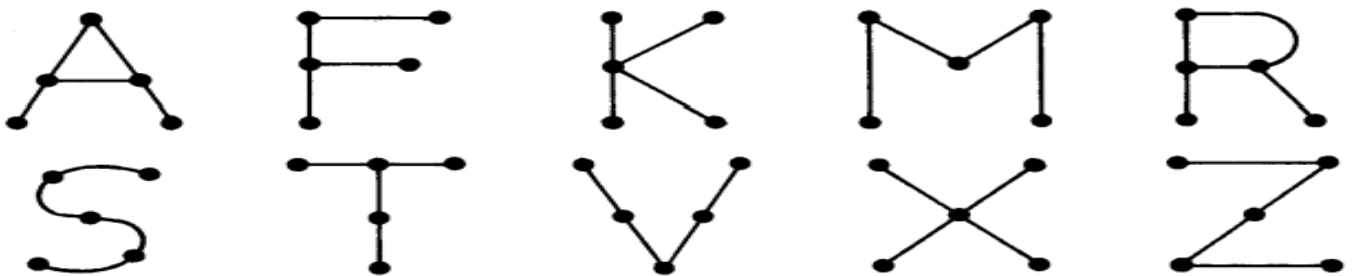
Function: $g(V_1) = U_5$, $g(V_2) = U_4$ $g(V_3) = U_3$ $g(V_4) = U_2$
 $g(V_5) = U_7$ $g(V_6) = U_1$ $g(V_7) = U_6$

d)



Solution: Graph G has no vertex of degree 4 where G' has vertex V_2 with degree 4. Hence, they are not isomorphic.

26. Given below are different graphs of Letters, identify which are the isomorphic to each other. Provide the invariant for graph isomorphism that they share or do not share.

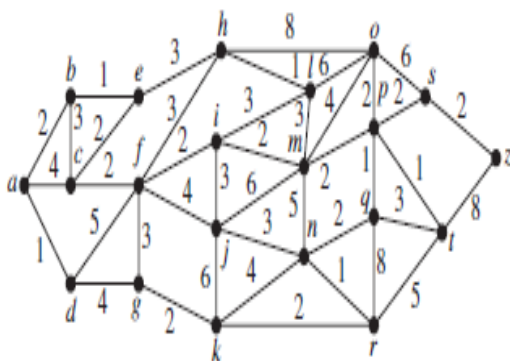


Solution:

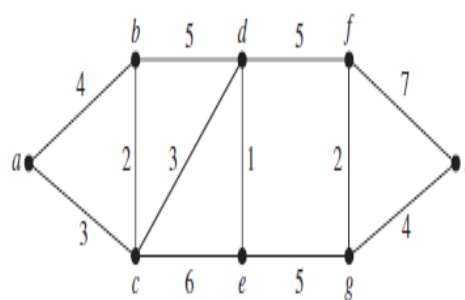
We note that A and R are isomorphic graphs. Also, F and T are isomorphic graphs, K and X are isomorphic graphs and M, S, V, and Z are isomorphic graphs.

27. Find the length of a shortest path between node a and node z in the given weighted graph by using Dijkstra's algorithm.

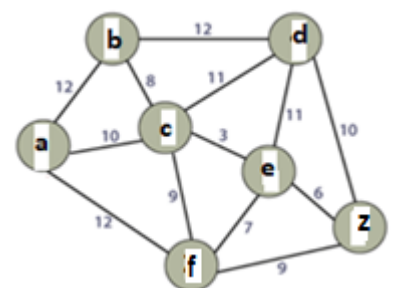
a)



b)



c)



a) Solution:

N	D(b)	D(c)	D(d)	D(e)	D(f)	D(g)	D(h)	D(i)	D(j)	D(k)	D(l)	D(m)	D(n)	D(o)	D(p)	D(q)	D(r)	D(s)	D(t)	D(z)
a	2,a	4,a	1,a																	
ad	2,a	4,a			6,d	5,d														
adb		4,a		3,b	6,d	5,d														
adbe		4,a			6,d	5,d	6,e													
adbec					6,c	5,d	6,e													
adbeg					6,c		6,e		7,g											
adbegf							6,e	8,f	10,f	7,g										
adbegfh								8,f	10,f	7,g	7,h			14,h						
adbegfhk								8,f	10,f		7,h		11,k	14,h			9,k			
adbegfhkl								8,f	10,f			10,l	11,k	13,l			9,k			
adbegfhkli									10,f			10,l	11,k	13,l			9,k			
adbegfhklir									10,f			10,l	10,r	13,l		17,r			14,r	
adbegfhklirj												10,l	10,r	13,l		17,r			14,r	
adbegfhklirjm													10,r	13,l	12,m	17,r			14,r	
adbegfhklirjmn														13,l	12,m	12,n			14,r	
adbegfhklirjmnp														13,l		12,n		14,p	13,p	
adbegfhklirjmnpq														13,l				14,p	13,p	
adbegfhklirjmnpqo																		14,p	13,p	
adbegfhklirjmnpqot																		14,p		21,t
adbegfhklirjmnpqots																				16,s
adbegfhklirjmnpqotsz																				

b) Solution:

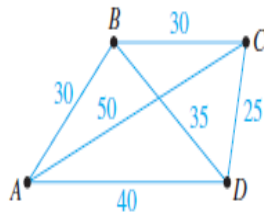
N	D(b)	D(c)	D(d)	D(e)	D(f)	D(g)	D(z)
a	4,a	3,a	∞	∞	∞	∞	∞
ac			6,c	9,c	∞	∞	∞
acb			6,c	9,c	∞	∞	∞
acbd				7,d	11,d	∞	∞
acbde					11,d	12,e	∞
acbdef						12,e	18,f
acbdefg							16,g
acbdefgz	4,a	3,a	6,c	7,d	11,d	12,e	16,g

c) Solution:

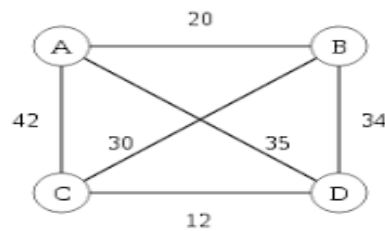
N	D(b)	D(c)	D(d)	D(e)	D(f)	D(z)
a	12,a	10,a	∞	∞	12,a	∞
ac	12,a		21,c	13,c	12,a	∞
acb			21,c	13,c	12,a	∞
acbf			21,c	13,c		21,f
acbfz			21,c			19,e
acbfz			21,c			
acbfz	12,a	10,a	21,c	13,c	12,a	19,f

28. Imagine that the drawing below is a map showing four cities and the distances in kilometers between them. Suppose that a salesman must travel to each city exactly once, starting and ending in city A. Which route from city to city will minimize the total distance that must be traveled?

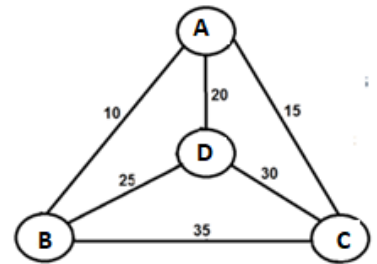
a)



b)



c)



i) Solution:

Hamiltonian Circuit are: ABCDA = 125;

ABDCA = 140;

ACBDA = 155.

Hence ABCDA = 125 is the minimum distance travelled.

ii) Solution:

Hamiltonian Circuit are: ABCDA = 97;

ABDCA = 108;

ACBDA = 141.

Hence ABCDA = 97 is the minimum distance travelled.

iii) Solution:

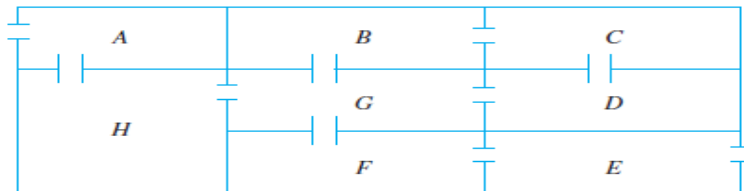
Hamiltonian Circuit are: ABCDA = 95;

ABDCA = 80;

ACBDA = 95.

Hence ABDCA = 80 is the minimum distance travelled.

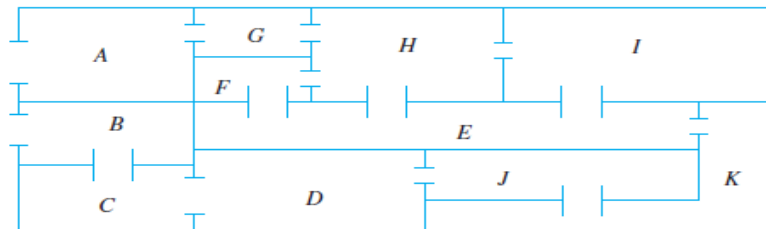
29. a) The following is a floor plan of a house. Is it possible to enter the house in room A, travel through every interior doorway of the house exactly once, and exit out of room E? If so, how can this be done?



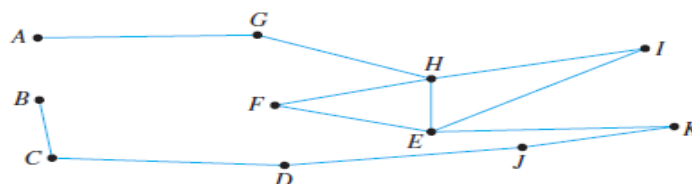
Solution:

Yes! Path: A → H → G → B → C → D → G → F → E

- b) The floor plan shown below is for a house that is open for public viewing. Is it possible to find a trail that starts in room A, ends in room B, and passes through every interior doorway of the house exactly once? If so, find such a trail.



Solution Let the floor plan of the house be represented by the graph below.



Each vertex of this graph has even degree except for A and B, each of which has degree 1. Hence by Corollary 10.2.5, there is an Euler path from A to B. One such trail is

AGHFEIHEKJDCB.

c) You are given a 10 pieces of domino set whose titles have the following set of dots:

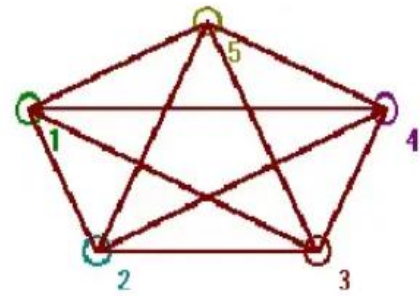
(1,2); (1,3); (1,4); (1,5); (2,3); (2,4); (2,5); (3,4); (3,5); (4,5).

Discuss the possibility of arranging the titles in a connect series such that number on a title always touches the same number on it neighbor. (Hint: Use a five vertex complete graph and see if it is an Euler Graph.)

Solution:

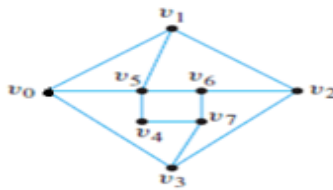
This is a Euler graph as 1-2-3-1-5-2-4-3-5-4-1

$\{(1,2), (2,3), (1,3), (1,5), (2,5), (2,4), (3,4), (3,5), (4,5), (1,4)\}$

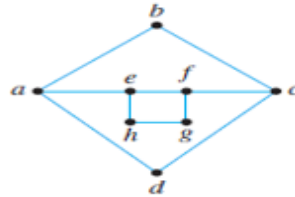


30. a) Find Hamiltonian Circuits and Path for those graphs that have them. Explain why the other graphs do not.

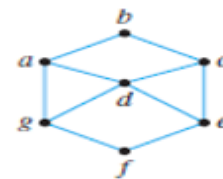
i)



ii)



iii)



i) Solution:

Hamiltonian Circuit: $V_0, V_1, V_2, V_6, V_5, V_4, V_7, V_3, V_0$.

Hamiltonian Path: $V_0, V_1, V_2, V_6, V_5, V_4, V_7, V_3$

ii) Solution:

Hamiltonian Circuit: doesn't exist.

Hamiltonian Path: b, c, f, g, h, e, a, d

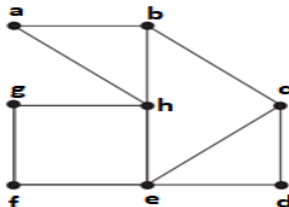
iii) Solution:

Hamiltonian Circuit: d, c, b, a, g, f, e, d .

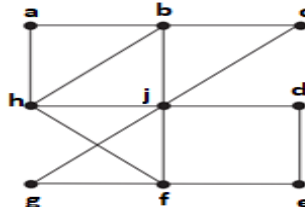
Hamiltonian Path: d, c, b, a, g, f, e

b) Determine which of the graphs have Hamiltonian circuits and Path. If the graph does not have Hamiltonian Circuits and Path, explain why not. If it does have Hamiltonian Circuit and Path, describe one.

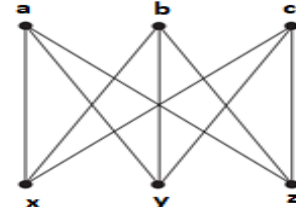
i)



ii)



iii)



Solution:

i) Solution:

Hamiltonian Circuit: $a, b, c, d, e, f, g, h, a$.

Hamiltonian Path: a, b, c, d, e, f, g, h .

ii) Solution:

Hamiltonian Circuit: doesn't exist.

Hamiltonian Path: doesn't exist

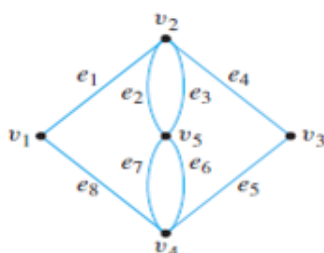
iii) Solution:

Hamiltonian Circuit: a, x, b, y, c, z, a .

Hamiltonian Path: a, x, b, y, c, z .

31. a) Determine which of the graphs have Euler circuits. If the graph does not have an Euler circuit, explain why not. If it does have an Euler circuit, describe one.

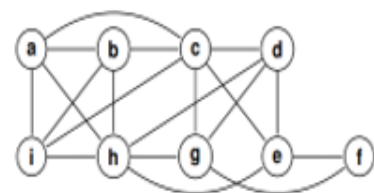
i)



ii)



iii)



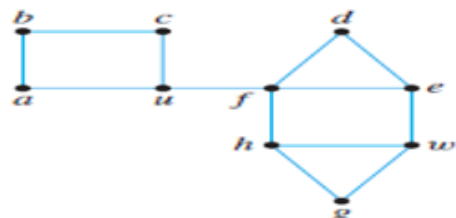
i) Solution: All vertices have even degree so circuit exists.

Euler Circuit: $V_1, V_2, V_5, V_4, V_5, V_2, V_3, V_4, V_1$

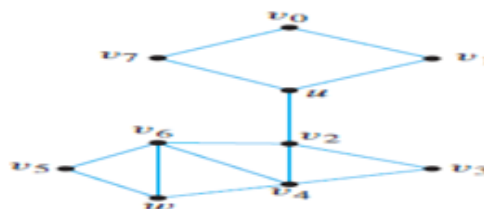
ii) Solution:

Euler Circuit: a, b, c, a, i, b, h, d, c, i, h, g, d, e, f, g, c, e, h, a.

i)



ii)



Euler path: U, V₁, V₀, V₇, U, V₂, V₃, V₄, V₂, V₆, V₅, W, V₆, V₄, W.

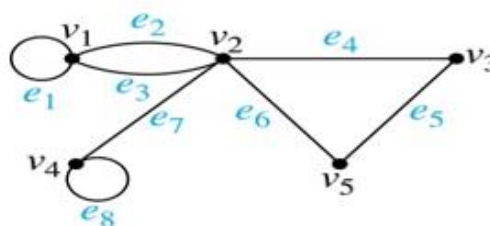
i)



Solution: i)

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

ii)



ii)

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

i)

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

i)



ii)

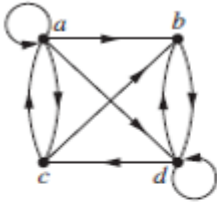
$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

ii)



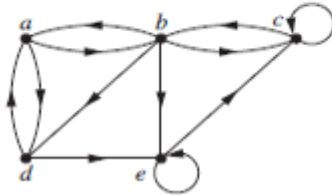
18

i)



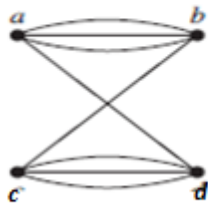
Initial Vertex	Terminal Vertices
a	a, b, c, d
b	d
c	a, b
d	b, c, d

ii)



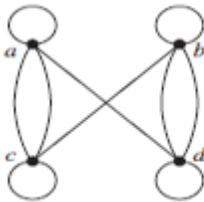
Initial Vertex	Terminal Vertices
a	b, d
b	a, c, d, e
c	b, c,
d	a, e
e	c, e

iii)



Vertex	Adjacent Vertices
a	b, d
b	a, c
c	b, d
d	a, c

iv)



Vertex	Adjacent Vertices
a	a, c, d
b	b, c, d
c	a, b, c
d	a, c, d

b) Draw the graph represented by the given adjacency matrix.

(i)

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

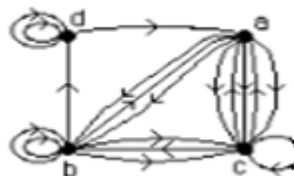
Solution:



(ii)

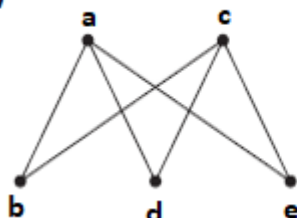
$$\begin{bmatrix} 0 & 2 & 3 & 0 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$$

Solution:

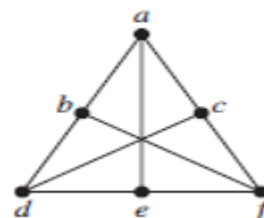


34. a) Determine whether the given graph is planar. If so, draw it so that no edges cross.

i)



ii)

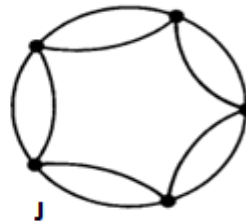
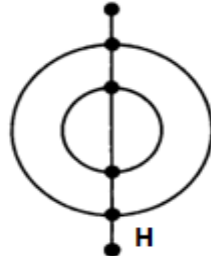


Solution:

- i) This graph can be untangled if we play with it long enough. The following picture gives a planar representation of it.
- ii) This graph is not planar because we can form a $K_{3,3}$ graph with the vertices $\{a, d, f\}$ and $\{b, c, e\}$.



- b) Count the number of vertices V , the number of edges E , and the number of regions R of each graph given below and also verify Euler's formula.

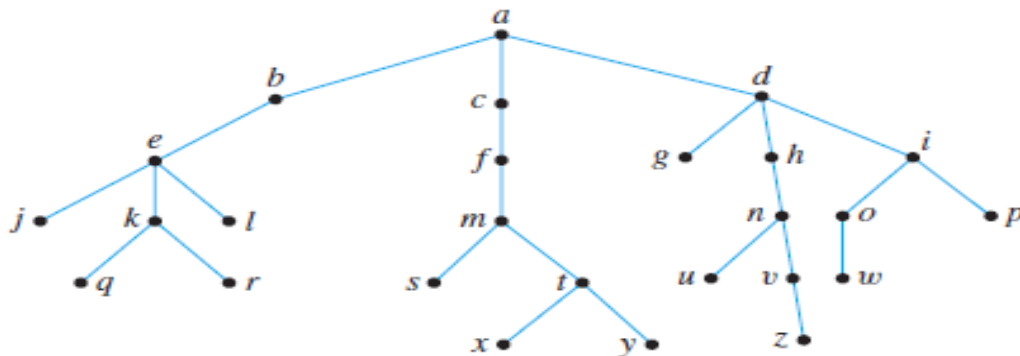


Solution: Graph G: $V = 4$, $E = 6$, $R = 4$. Hence, $R = E - V + 2 = 6 - 4 + 2 = 4$.

Solution: Graph H: $V = 6$, $E = 9$, $R = 5$; Hence, $R = E - V + 2 = 9 - 6 + 2 = 5$.

Solution: Graph J: $V = 5$, $E = 10$, $R = 7$. Hence, $R = E - V + 2 = 10 - 5 + 2 = 7$.

35. Consider the tree shown at right with root a.



Solution:

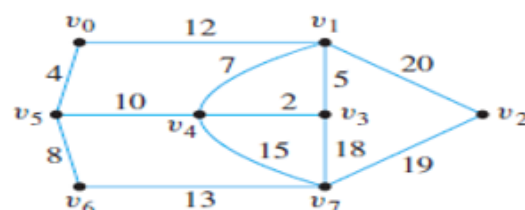
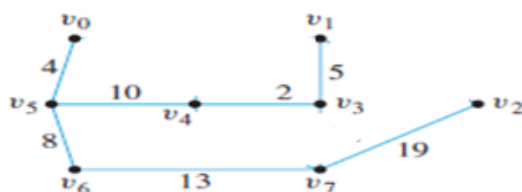
- i) What is the level of n ?
- ii) What is the level of a ?
- iii) What is the height of this rooted tree?
- iv) What are the children of n ?
- v) What is the parent of g ?
- vi) What are the siblings of j ?
- vii) What are the descendants of f ?
- viii) What are the internal nodes?
- ix) What are the ancestors of z ?
- x) What are the leaves?

Level of n is 3.
 Level of a is 0.
 Height of this rooted tree is 5
 u & v are the children of n .
 d is the parent of g .
 k & l are the siblings of j .
 m , s , t , x & y are the descendants of f .
 a , b , e , k , c , f , m , t , d , h , i , n , o & v are the internal nodes.
 v , n , h , d & a are the ancestors of z .
 j , l , q , r , s , x , y , g , p , u , w & z are the leaves.

36. a) Use Prim's algorithm to find a minimum spanning tree starting from V_0 for given graphs. Indicate the order in which edges are added to form each tree.

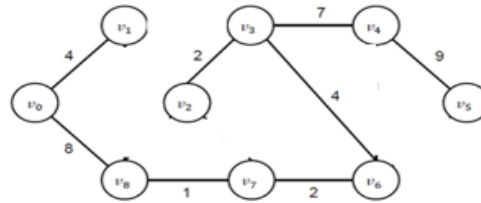
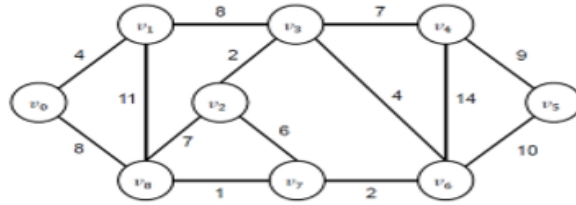
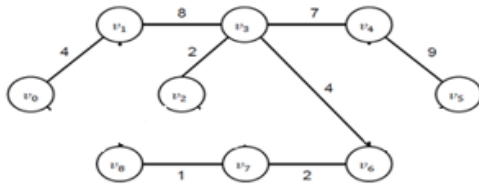
i) Solution: MST Cost = 61

$(V_0, V_5) = 4$, $(V_5, V_6) = 8$, $(V_4, V_5) = 10$,
 $(V_3, V_4) = 2$, $(V_1, V_3) = 5$, $(V_6, V_7) = 13$,
 $(V_2, V_7) = 19$.



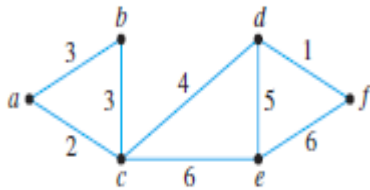
ii) Solution: MST Cost = 37

$(V_0, V_1) = 4$, $(V_0, V_8) = 8$, $(V_7, V_8) = 1$,
 $(V_6, V_7) = 2$, $(V_3, V_6) = 4$, $(V_2, V_3) = 2$,
 $(V_3, V_4) = 7$, $(V_4, V_5) = 9$,

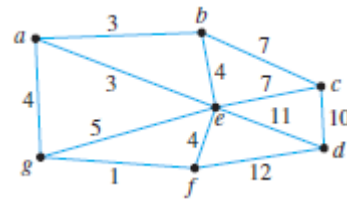


b) Use Kruskal's algorithm to find a minimum spanning tree for given graphs. Indicate the order in which edges are added to form each tree.

i)



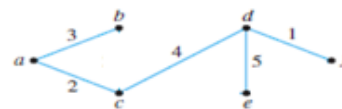
ii)



i) Solution: MST cost = 15

Order of edges added is:

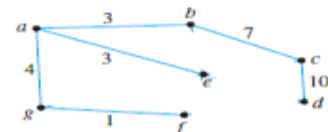
$(d, f)=1$, $(a, c)=2$, $(a, b)=3$, $(b, e)=3$,
 $(c, d)=4$, $(d, e)=5$, $(e, f)=6$, $(e, g)=6$



ii) Solution: MST cost = 28

Order of edges added is:

$(g, f)=1$, $(a, b)=3$, $(a, e)=3$, $(a, g)=4$,
 $(b, e)=4$, $(e, f)=4$, $(g, e)=5$, $(b, c)=7$,
 $(e, g)=7$, $(c, d)=10$, $(d, e)=11$, $(d, f)=12$



37. (a) Build a binary search tree for the give set of word's using alphabetical order:

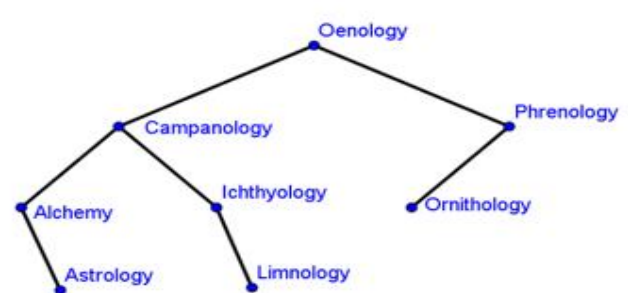
i) banana, peach, apple, pear, coconut, mango, and papaya.

ii) oenology, phrenology, campanology, ornithology, ichthyology, limnology, alchemy, and astrology.

i)



ii)



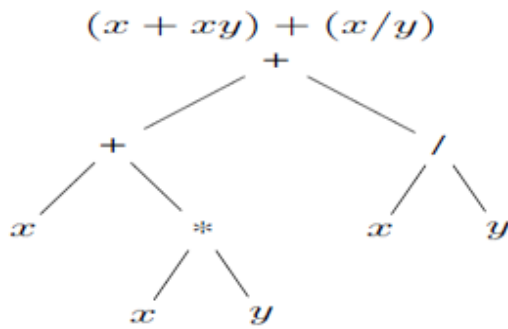
(b) Represent these expressions using binary trees.

(i) $(x + xy) + (x / y)$

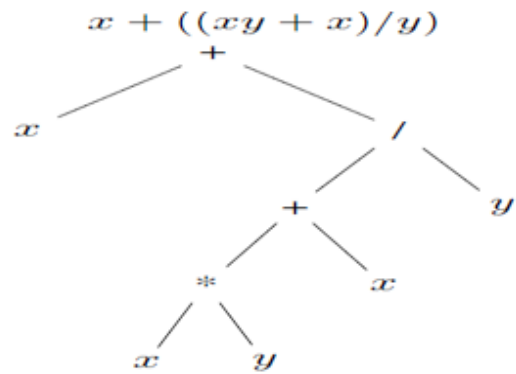
Solution:

(ii) $x + ((xy + x) / y)$

i)

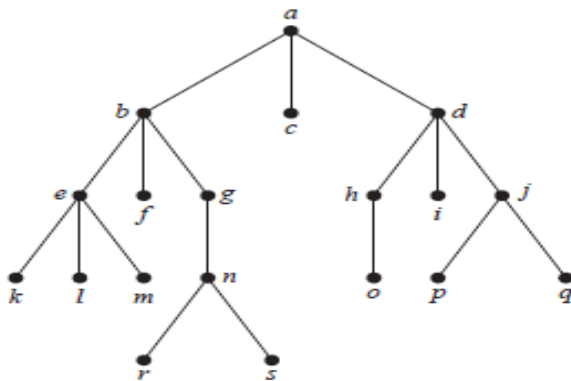


ii)

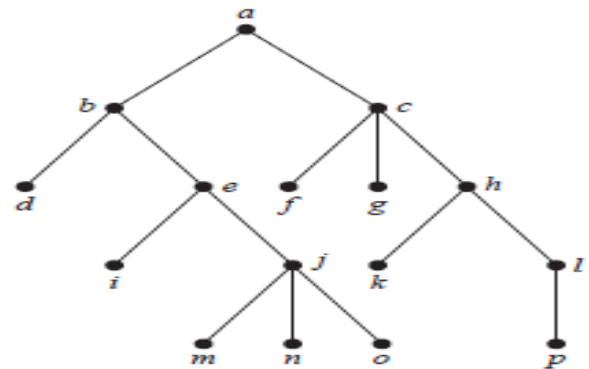


38. a) Determine the order in which Preorder, Inorder and Postorder traversal visits the vertices of the given ordered rooted tree.

i)



ii)



Solution:

i)

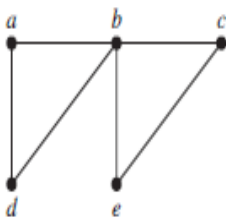
Pre order: a b e k l m f g n r s c d h o i p q
 In order: k e l m b f r n s g a c o h d i p j q
 Post order: k l m e f r s n g b c o h l p q j d a

ii)

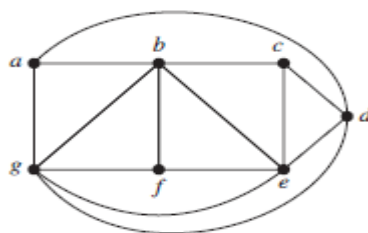
Pre order: a b d e i j m n o c f g h k l p
 In order: d b i e m j n o a f c g k h p l
 Post order: d i m n o j e b f g k p l h c a

b) Find a spanning tree for the graph shown by removing edges in simple circuits. Write down the removed edges.

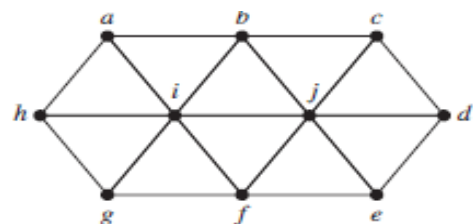
i)



ii)

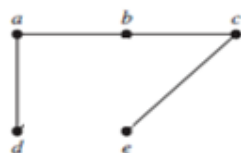


iii)

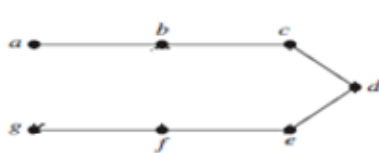


Solution:

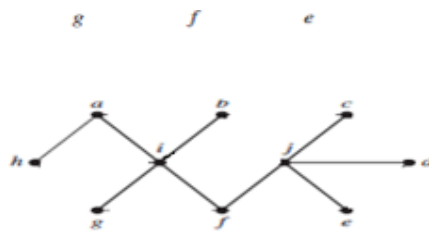
i)



ii)



iii)



39. a) How many edges does a tree with 10000 vertices have?

Solution:

A tree with n vertices has $n - 1$ edge. Hence $10000 - 1 = 9999$ edges.

(b) How many edges does a full binary tree with 1000 internal vertices have?

Solution:

A full binary tree has two edges for each internal vertex. So, we'll just multiply the number of internal vertices by the number of edges. Hence $1000 * 2 = 2000$ edges.

(c) How many vertices does a full 5-ary tree with 100 internal vertices have?

Solution:

A full m -ary tree with I internal vertices has $n = mi + 1$ vertices.

From the given information, we have $m = 5$, $i = 100$

So $n = 5 \times 100 + 1 = 501$

Therefore a full 5-ary tree with 100 internal vertices has 501 vertices.

40. (a) Write these expressions in Prefix and Postfix notation:

i) $(x + xy) + (x / y)$

Solution:

Prefix: $++x*xy/xy$

Postfix: $xx y * + xy / +$

ii) $x + ((xy + x) / y)$

Solution:

Prefix: $+x/+*xyxy$

Postfix: $xx y * x + y / +$

b) i) What is the value of this prefix expression $+ - \uparrow 3 2 \uparrow 2 3 / 6 - 4 2$

Solution: 4

ii) What is the value of this postfix expression $4 8 + 6 5 - * 3 2 - 2 2 + * /$

Solution: 3

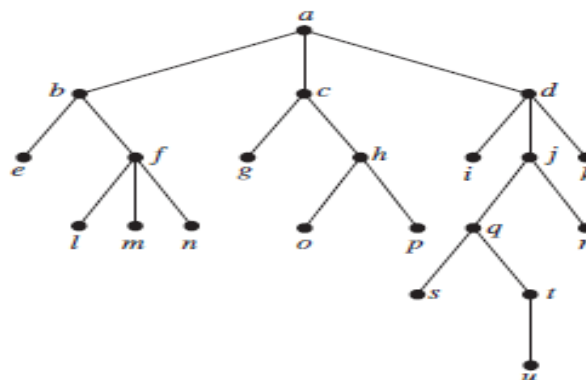
41. Answer these questions about the rooted tree illustrated.

a) Is the rooted tree a full m-ary tree?

Solution: It is not a full m-ary tree for any m because some of its internal vertices have two children and others have three children.

b) Is the rooted tree a balanced m-ary tree?

Solution: It is not balanced m-ary tree because it has leaves at levels 2, 3, 4 and 5.



c) Draw the subtree of the tree that is rooted at

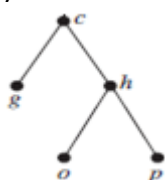
a) c.

b) f.

c) q.

Solution:

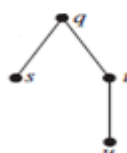
a)



b)



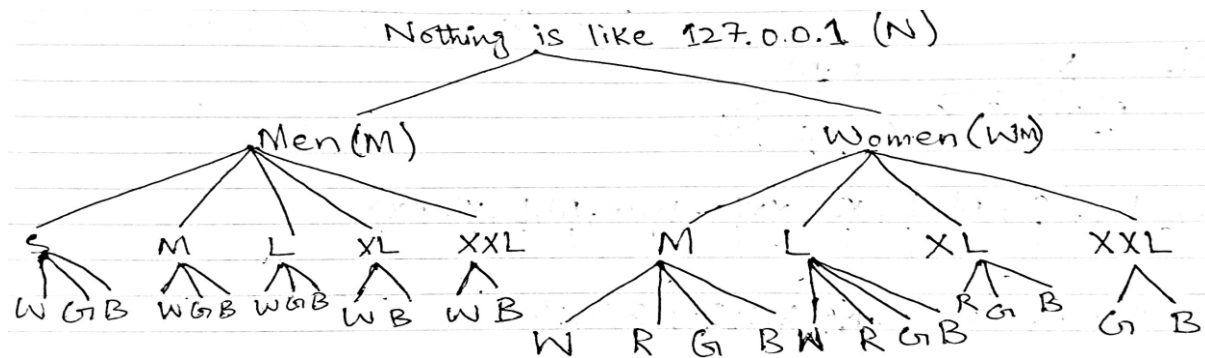
c)



42. Suppose that a popular company launches "Nothing is like 127.0.0.1" for both men and women. Women T-shirts come in four different sizes: M, L, XL, and XXL. Each size comes in four colors (white, red, green, and black), except XL, which comes only in red, green, and black, and XXL, which comes only in green and black. Men T-shirts come in five different sizes: S, M, L, XL, and XXL. Each size comes in three colors (white, green, and black), except XL and XXL, which comes only in white and black.

a) Use a tree diagram to determine what is the minimum number of T-shirts that the shop needs to stock to have one of each size and color available for both men and women?

Solution: 26 T-shirts needed to be stocked.



b) Consider the tree obtained in part (a).

Show how many internal vertices and leaves the tree has? Determine the height of the tree?

Solution:

Internal vertices: 12, Leaves: 26, Height: 03

c) Consider the tree obtained in part (a).

i) Determine whether it is a Full m-ary tree or not? Give reason.

Solution:

It is not full m-ary trees all the internal vertices have different no. of children.

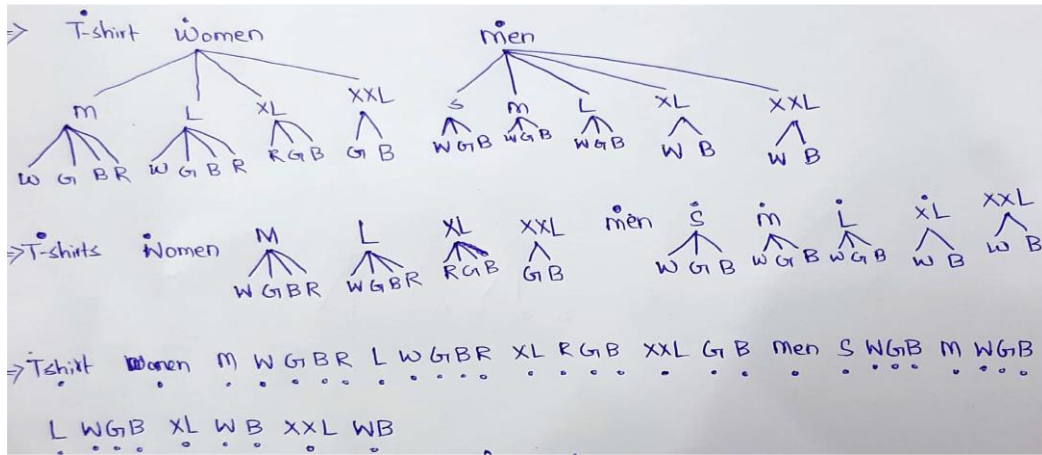
ii) Determine whether it is a balanced m-ary tree or not? Give reason.

Solution:

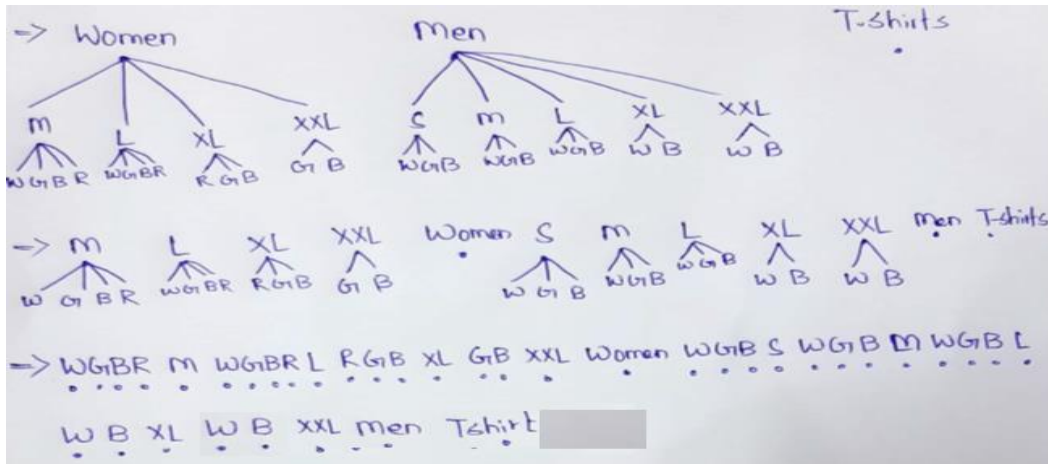
It is balanced m-ary trees as all the leaves are at h- level.

d) Show the Preorder and Post order traversal of the tree obtained in part (a).

Solution: Pre order:



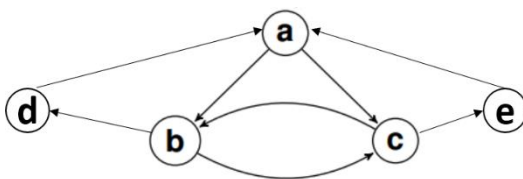
Post order:



43. Consider a scenario from World War II where two German soldiers are communicating with each other through a secure channel. Head of Platoon want to communicate two different messages to each of them through separate communication channels. Both of them can received the message but can't reply back. After receiving the message from the head of platoon, these two soldiers communicate the received message to one of their subordinates each. These subordinates can't reply to them but have a one-way communication channel with the Platoon Head so that they can confirm the message delivery to the head of platoon.

a) Transform the given scenario into a digraph. Show the in-degree and out-degree of each vertex.

Solution:



In Degree: $\deg^-(a) = 2$, $\deg^-(b) = 2$, $\deg^-(c) = 2$, $\deg^-(d) = 1$, $\deg^-(e) = 1$

Out Degree: $\deg^+(a) = 2$, $\deg^+(b) = 2$, $\deg^+(c) = 2$, $\deg^+(d) = 1$, $\deg^+(e) = 1$

b) Using diagram in part a), determine whether the relation is an equivalence relation OR a partial-order relation? Show all of your steps.

Solution:

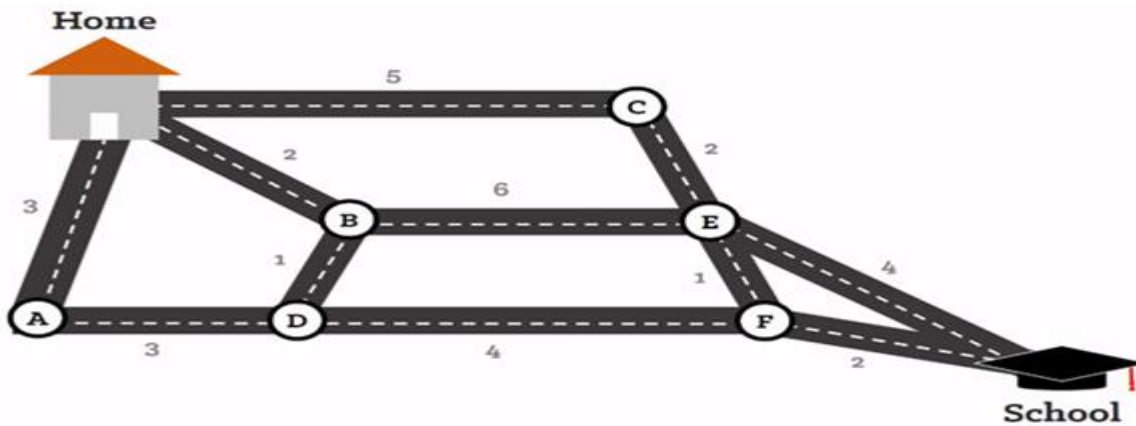
For equivalence relation, relation should be reflexive, Symmetric and Transitive.

For partial Order relation, relation should be reflexive, Antisymmetric and Transitive.

Relation	Reason
Not reflexive	Because every vertex does not have a self-loop.
Not symmetric	Because we do not have (b, a) for (a, b) .
Not antisymmetric	Because we have (b, c) and (c, b) .
Not transitive	Because the path for (b, c) , (c, b) from b to b is missing the edge (b, b) .

Hence, the relation is neither Equivalence nor Partial order relation.

44. Musab is an undergraduate student. The below graph map shows different path from his home to the school.



a) Musab wants to determine the shortest path from home to school using Dijkstra Algorithm. What will be the cost of that path? Show all steps.

Solution:

The shortest path, which could be found using Dijkstra's algorithm, is Home $\rightarrow$ B $\rightarrow$ D $\rightarrow$ F $\rightarrow$ School

Cost: $2 + 1 + 4 + 2 = 9$.

N	D(A)	D(B)	D(C)	D(D)	D(E)	D(F)	D(S)
H	3,H	2,H	5,H	∞	∞	∞	∞
HB	3,H		5,H	3,B	8,B	∞	∞
HBA			5,H	3,B	8,B	∞	∞
HBAD			5,H		8,B	7,D	∞
HBADC					7,C	7,D	∞
HBADCE						7,D	11,E
HBADCEF							9,F
HBADCEFS							

b) Determine the Euler Circuit and Path in the above graph for Musab? Show all of your steps with reasoning.

Solution:

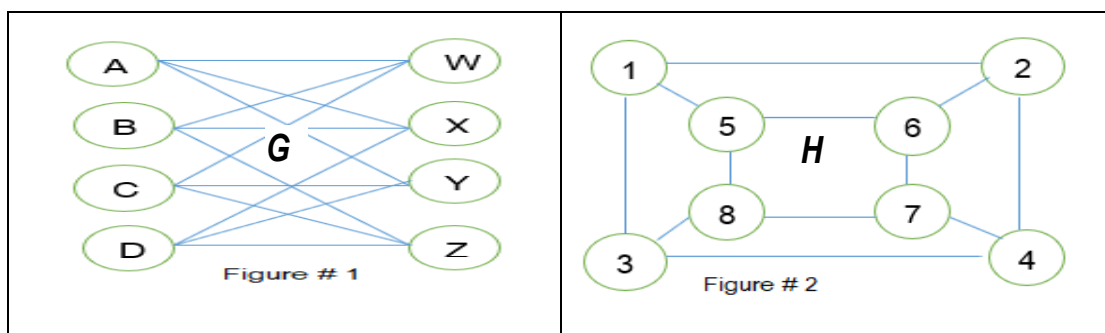
No Euler Circuit possible because some nodes have odd vertices. No Euler path is possible because there are more than two vertices with odd degree.

c) Determine the Hamilton Circuit and Path in the above graph for Musab? Show all of your steps with reasoning.

Solution:

No Hamilton Circuit But path is possible. E.g: C $\rightarrow$ E $\rightarrow$ S $\rightarrow$ F $\rightarrow$ D $\rightarrow$ B $\rightarrow$ H $\rightarrow$ A

45. a) Determine if the following two graphs G and H as shown in figure# 1 & 2 are isomorphic. If they are, give function $f: V(G) \rightarrow V(H)$ that define the isomorphism. If they are not, give the reason why?



Solution:

Both Graph G and H are isomorphic. $f(A) = 1, f(B) = 6, f(C) = 8, f(D) = 3, f(W) = 5, f(X) = 2, f(Y) = 4, f(Z) = 7$

b) Consider the graphs G and H from part (a).

i) Is Graph G a complete Bipartite Graph? Explain.

Solution:

Graph G is not complete bipartite as A is not adjacent with Z , B is not adjacent with Y , C is not adjacent with X , and D is not adjacent with W .

ii) Determine if Euler and Hamilton circuits exist in Graph H . If yes, show the circuit, if not explain why?

Solution:

Graph H doesn't contain Euler Circuit as all the vertices have odd degree.

Graph H Contains Hamilton's Circuits:

H1: 1,2,3,7,6,5,8,4,1

H2: 1,2,6,7,3,4,8,5,1

H3: 1,2,6,5,8,7,3,4,1

46. Let $A = \{a, b, c, d\}$ and $B = \{a, b, c, d\}$. Consider the following functions:

a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$

b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$

c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$

d) $f(a) = c, f(b) = a, f(c) = b, f(d) = d$

(i) Determine the Domain, Co-domain and Range of the functions.

Solution: a) Domain: $\{a, b, c, d\}$ Co-domain: $\{a, b, c, d\}$ Range: $\{a, b, c, d\}$

b) Domain: $\{a, b, c, d\}$ Co-domain: $\{a, b, c, d\}$ Range: $\{b, c, d\}$

c) Domain: $\{a, b, c, d\}$ Co-domain: $\{a, b, c, d\}$ Range: $\{b, c, d\}$

d) Domain: $\{a, b, c, d\}$ Co-domain: $\{a, b, c, d\}$ Range: $\{a, b, c, d\}$

(ii) Determine whether the functions are Injective, Surjective and Bijective or not?

a) Injective and Bijective

b) Neither

c) Neither

d) Injective and Bijective

(iii) Determine the inverse of function if exists.

a) $f^{-1}(b) = a, f^{-1}(a) = b, f^{-1}(c) = c, f^{-1}(d) = d$

b) Not exists

c) $f^{-1}(c) = a, f^{-1}(a) = b, f^{-1}(b) = c, f^{-1}(d) = d$

d) Not exists

47. a) Let $f(x) = \left\lfloor \frac{x^2}{3} \right\rfloor$, Find $f(S)$ if :

(i) $S = \{-2, -1, 0, 1, 2, 3\}$ Solution: $f(S) = \{1, 0, 0, 0, 1, 3\}$

(ii) $S = \{0, 1, 2, 3, 4, 5\}$ Solution: $f(S) = \{0, 0, 1, 3, 5, 8\}$

(iii) $S = \{1, 5, 7, 11\}$ Solution: $f(S) = \{0, 8, 16, 40\}$

(iv) $S = \{2, 6, 10, 14\}$ Solution: $f(S) = \{1, 12, 33, 65\}$

b) (i) $\left\lceil \frac{3}{4} \right\rceil$ Solution: 1

(ii) $\left\lfloor \frac{7}{8} \right\rfloor$ Solution: 0

iii) $\left\lceil -\frac{3}{4} \right\rceil$ Solution: 0

(iv) $\left\lfloor -\frac{7}{8} \right\rfloor$ Solution: -1

(v) $\lceil 3 \rceil$ Solution: 3

(vi) $\lfloor -1 \rfloor$ Solution: -1

(vii) $\left\lfloor \frac{1}{2} + \left\lceil \frac{3}{2} \right\rceil \right\rfloor$ Solution: 2

(viii) $\left\lceil \frac{1}{2} \cdot \left\lfloor \frac{5}{2} \right\rfloor \right\rceil$ Solution: 1

c) Prove or disproof that if x is a real number, then $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$.

Solution: $\left\lfloor -\frac{3}{4} \right\rfloor = -\left\lceil \frac{3}{4} \right\rceil \rightarrow -1 = -1$

$\left\lceil -\frac{3}{4} \right\rceil = -\left\lfloor \frac{3}{4} \right\rfloor \rightarrow 0 = 0$

Hence, It's a disproof.

48. Let f and g be the functions from the set of integers to the set of integers defined by $f(a) = 2a + 3$ and $g(a) = 3a + 2$.

a) What is the composition of f and g ? What is the composition of g and f ?

Solution: $(f \circ g)(a) = f(g(a)) = f(3a + 2) = 2(3a + 2) + 3 = 6a + 7$.

$(g \circ f)(a) = g(f(a)) = g(2a + 3) = 3(2a + 3) + 2 = 6a + 11$.

b) Which type of function f and g are?

Solution: f and g are not injective, surjective and bijective.

c) Are f and g invertible?

Solution: Hence f and g are not invertible.